```
# 1.Load the Tips dataset into a DataFrame.
import pandas as pd
import numpy as np
import seaborn as sns
```

```
df=sns.load_dataset('tips')
```

```
df
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

Next steps: Generate code with df | New interactive sheet

```
# 2.Display the first five rows.
df.head(5)
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: Generate code with df | New interactive sheet

```
#3. Check the data types of all columns using df.dtypes.
```

```
df.dtypes
```

	0
total_bill	float64
tip	float64
sex	category
smoker	category
day	category
time	category
size	int64

dtype: object

```
# 4 Filter customers whose total_bill is greater than 20.
result=df[df['total_bill']>20]
```

```
print(result)
```

```
print(result)
```

```
     total_bill   tip     sex smoker  day    time  size
2         21.01  3.50    Male     No  Sun  Dinner     3
3         23.68  3.31    Male     No  Sun  Dinner     2
4         24.59  3.61  Female     No  Sun  Dinner     4
5         25.29  4.71    Male     No  Sun  Dinner     4
7         26.88  3.12    Male     No  Sun  Dinner     4
..          ...   ...     ...    ...  ...     ...   ...
237       32.83  1.17    Male    Yes  Sat  Dinner     2
238       35.83  4.67  Female     No  Sat  Dinner     3
239       29.03  5.92    Male     No  Sat  Dinner     3
240       27.18  2.00  Female    Yes  Sat  Dinner     2
241       22.67  2.00    Male    Yes  Sat  Dinner     2

[97 rows x 7 columns]
```

```
#5. Explain why size is numerical but day is categorical.
size – Numerical and discrete
- It represents the number of people at a table.
- It contains countable whole numbers such as 1, 2, 3, and 4.
- It is discrete numerical data because people are counted in whole numbers.

day – Categorical and nominal
- It represents categories such as Thursday, Friday, Saturday, and Sunday.
- These values identify groups rather than measurable quantities.
- It is nominal categorical data because the days in this dataset are labels, not numerical measurements or a ranked scale.
```

Start coding or generate with AI.